

ClouDrive™: Autonomy with Real Human Control

ClouDrive™ is the ability to operate a vehicle remotely, with a human driver in full control, using real-time video, audio, and sensor feedback from the vehicle.

At Turtle Mobility, tele-driving is not a backup feature; it is a **core capability**. It allows users to either drive the vehicle themselves or hand over control to a certified remote driver whenever they choose. This delivers many of the benefits of autonomous driving, while keeping **human judgment in the loop**-especially important in complex, real-world urban environments.

Ultra-Low Latency Connectivity

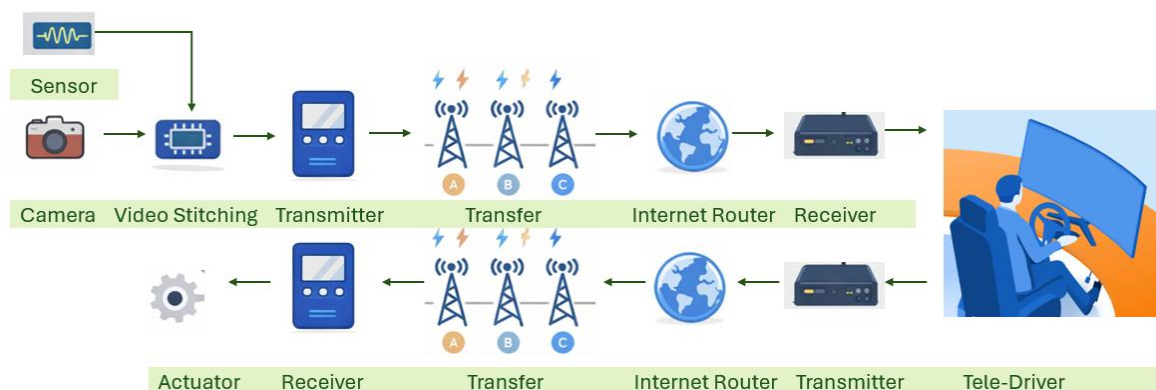
Advancements in mobile networks have made safe and responsive tele-driving possible at scale.

- With **5G**, we achieve near real-time video streaming with **extremely low end-to-end latency**
- Even on **modern 4G networks**, we consistently operate with **glass-to-glass video latency below ~60 milliseconds**

This low latency enables remote drivers to:

- React naturally and instinctively
- Perform precise maneuvers
- Drive smoothly and confidently in dense urban traffic

The result is a driving experience that feels **immediate, predictable, and human**, not delayed or robotic.

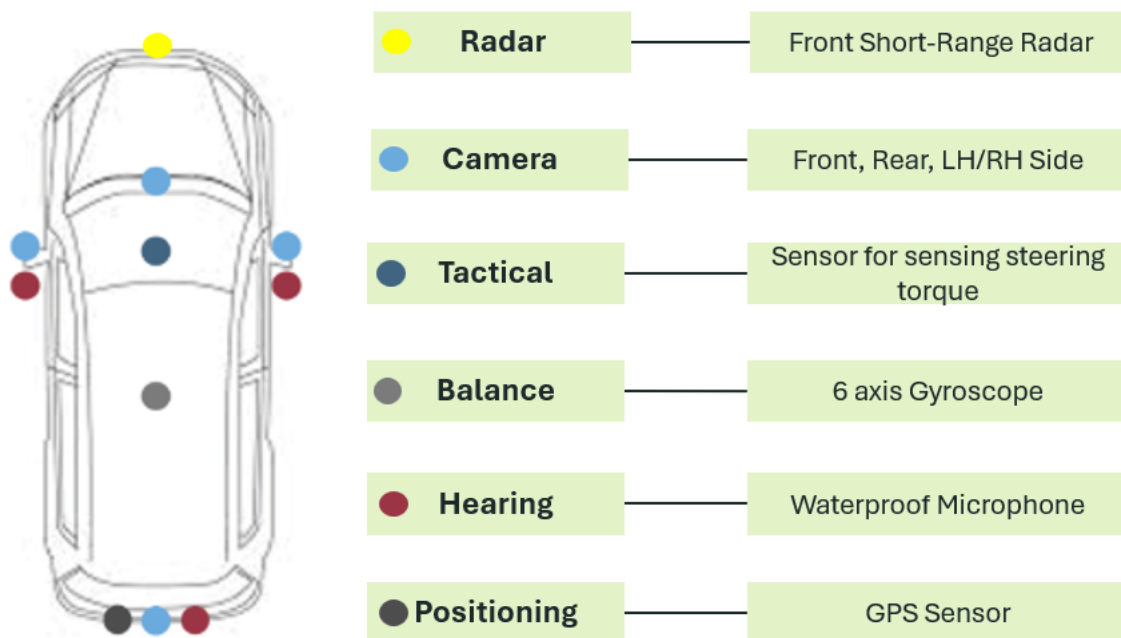


Minimal, Purpose-Driven Sensor Architecture

True to our KiSS design DNA, Turtle Mobility uses a **minimal yet safety-focused sensor array**- only what is truly required to drive safely and reliably.

Our core sensing stack includes:

- **Four cameras**, strategically placed to provide full situational awareness
- **Inertial sensors** (gyro) for vehicle motion and stability
- **GPS** for positioning and localization
- **Tactile feedback sensors** for vehicle dynamics and surface response
- **Audio sensing**, enabling environmental awareness and interaction



By deliberately avoiding unnecessary sensor redundancy, we:

- Reduce system complexity
- Improve reliability
- Lower cost
- Increase maintainability

Simplicity, when engineered correctly, is safer.

Multi-Network Connectivity

Vehicles are connected to multiple cellular networks at once, ensuring stable, real-time control even if one network experiences issues.

Radar-Based Safety Redundancy

While we keep the sensor stack minimal, **safety always comes first**.

For critical edge cases, Turtle Mobility includes:

- **Front-facing radar** as an independent redundancy layer

This radar system enables:

- **Autonomous emergency braking**
- **MRM (Minimum Risk Maneuver)** autonomously.
- Vehicle-initiated intervention **without dependence on the remote driver**
- Reliable obstacle detection under poor visibility conditions and giving drivers, preemptive alerts

This approach ensures that even in unexpected scenarios, the vehicle can take immediate protective action- adding a critical layer of safety without excessive complexity.

Radar-based MRM ensures controlled risk mitigation, even under degraded conditions like poor visibility.

ADAS Support for Remote Driving

Remote drivers are supported by a suite of **Advanced Driver Assistance Systems (ADAS)** designed to reduce cognitive load and enhance situational awareness.

These systems include:

- **Lane Departure Warning**
- **Lane Change Support**
- **Parking Assistance**
- **Emergency Braking Support**

ADAS functions act as **intelligent co-drivers**, assisting the human; not replacing them. This makes remote driving safer, less tiring, and more consistent, especially across long operating hours.

Software-Defined, Drive-by-Wire Functionality

Turtle Mobility's tele-driving capabilities are built on a **fully steer-by-wire and brake-by-wire architecture**.

This allows:

- Precise electronic control of motor torque, steering and braking
- Very low actuation latency
- Fewer mechanical components
- More interior space and design freedom

A software-defined platform enables continuous improvement and adaptability as technology and regulations evolve.

Technology With Intent

We do not chase maximum autonomy.

We build **practical autonomy**.

Our technology philosophy is simple:

- Use advanced connectivity where it adds real value
- Keep the system understandable and controllable
- Retain human decision-making where it matters most
- Design for real cities, real roads, and real people

Transparency for Occupants

A **cabin display** shows the **remote driver in control** while the vehicle is being tele-driven. This transparency reassures occupants that a human is actively engaged — creating trust and a stronger sense of safety.